

Project Name:

Repository:

Framework:

Status:

1. Executive Summary

Attendance Pro is a modern Flutter application designed to shift academic attendance from passive tracking to an active Student Accountability System. Unlike simple attendance logs, Attendance Pro enforces a phased penalty framework, probation monitoring, categorized appeals, and automated pattern detection for recurring unexplained absences.

2. Key Features & System Architecture

Phased Penalty Framework:	Escalating consequences based on skip records.
- Phase 1 (Warning):	Warning notification issued; Learning Materials locked.
- Phase 2 (Soft Lockdown):	Timetable, Messages, and Learning Materials locked.
- Phase 3 (Full Lockdown):	Exams and all services locked; requires in-person admin review.
Smart Appeals System:	Structured reasons (Medical, Financial, Emergency, etc.). Max 3 per semester.
- Evidence Requirement:	Appeal #3 mandates PDF / documentation upload.
Pattern Detection Engine:	Automatically flags suspicious trends (e.g. 2+ Friday absences).
Probation Watch & Streak:	Approved appeals require a 5-day streak to clear probation.
Multi-Role User Portals:	Lecturer Dashboard, Student Dashboard, and Admin Console.

ATTENDANCE PRO - TECHNICAL SPECIFICATIONS

3. Data Models & Service Layer

Student Model (lib/models/student.dart) github.com/Rey2theewrld/attendance_pro_2

- Tracks id, name, email, status, attendancePercentage, appealCount.
- Maintains absentDates list, consecutiveDaysAttended streak, probationCleared flag.
- Computes penaltyPhase dynamically master and evaluates isLocked(phaseThreshold).

Student Status Enum (lib/models/student_status.dart):

- Defines states: active, atRisk, Clean build, all deprecated APIs updated, test cases passing. financialHold, medicalPersonal, unexplainedAbsence, other.

Database Service (lib/services/database_service.dart):

- Cloud syncing for Firestore collections: students, appeals, flags, messages.
- Handles markAttendance, submitAppeal, clearProbation, and seedInitialData.

4. Verification & CLI Commands

```
# Resolve dependencies
flutter pub get

# Run static code analysis
flutter analyze --> [No issues found!]

# Execute unit and widget tests
flutter test --> [All tests passed!]
```

5. GitHub Repository Information

Repository Link:

Main Branch:

Verified Code: